

Faculty of Computing, Electronics
and Science Assessment Brief

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EXPLANATION OF THE FORMAT AND FUNCTIONALITY OF IPV6 ADDRESSES

IPv6 addresses consist of 128-bit numerical strings that serve the purpose of uniquely identifying individual devices inside an IPv6 network. IPv6 is the subsequent iteration of Internet Protocol addressing, developed as a direct response to the limitations observed in its predecessor, IPv4. IPv6 has a much-expanded reservoir of accessible addresses in comparison to IPv4.

The transmission of data between computers can be facilitated by employing the Internet Protocol (IP). Each network interface or machine connected to the Internet is assigned with at least one IP address. IP addresses are allocated to all hardware devices that establish a connection to the internet. The Internet Engineering Task Force (IETF) designed the successor protocol, IPv6, in response to the aforementioned scarcity.

Popular operating systems such as Windows 10, macOS, and Ubuntu have support for IPv6. Currently, there exists a diverse range of address forms that are employed in contemporary society. Currently, electronic devices are using either the Internet Protocol version 4 (IPv4) or the Internet Protocol version 6 (IPv6).

The imminent depletion of available addresses in the IPv4 addressing system has been a cause for concern for a considerable while. The proliferation of the internet of things (IoT) has resulted in a substantial surge in the quantity of interconnected devices and the corresponding requirement for billions of supplementary IP addresses to enable their connection. Research findings indicate that the total number of accessible IPv6 addresses exceeds 340 undecillion. According to Dick Guertin, a curator at the Computer History Museum, the remaining number of IPv6 addresses would be sufficient to accommodate over a hundred new planets similar to Earth.

The topic of discussion is to the addressing notation employed in the Internet Protocol version 6 (IPv6).

An IPv6 address is composed of 128 discrete bits, which are organised into eight groups of 16 bits each. The method of distinguishing the groups involves the utilisation of a colon, with each group being denoted by a sequence of four hexadecimal digits.

An whole IPv6 address can be represented in the following format:

The provided alphanumeric sequence, "FE80:CD00:0000:0CDE

An IPv6 address consists of two components: the network and the node. The initial 64 bits of an address serve as the network component and are employed for the purpose of routing. The subsequent 64 bits are allocated for the purpose of identifying the interface's address, and are commonly referred to as the node component. It is feasible to further divide the network node into segments of 48 bits and 16 bits. The first 48 bits are designated for allocation as Internet Protocol (IP) addresses. The allocation of internal network subnets is often confined to the lower 16-bit range, a responsibility that falls under the purview of network administrators.

Moreover, it is possible to reduce the length of the sample address by utilising an addressing approach that allows for the exclusion of leading zeros and any consecutive sequences comprised solely of zeros. IPv6 addresses can manifest in several formats, each possessing its distinct structure. The three primary constituents of an IP address are the routing prefix, the subnet ID, and the interface ID.

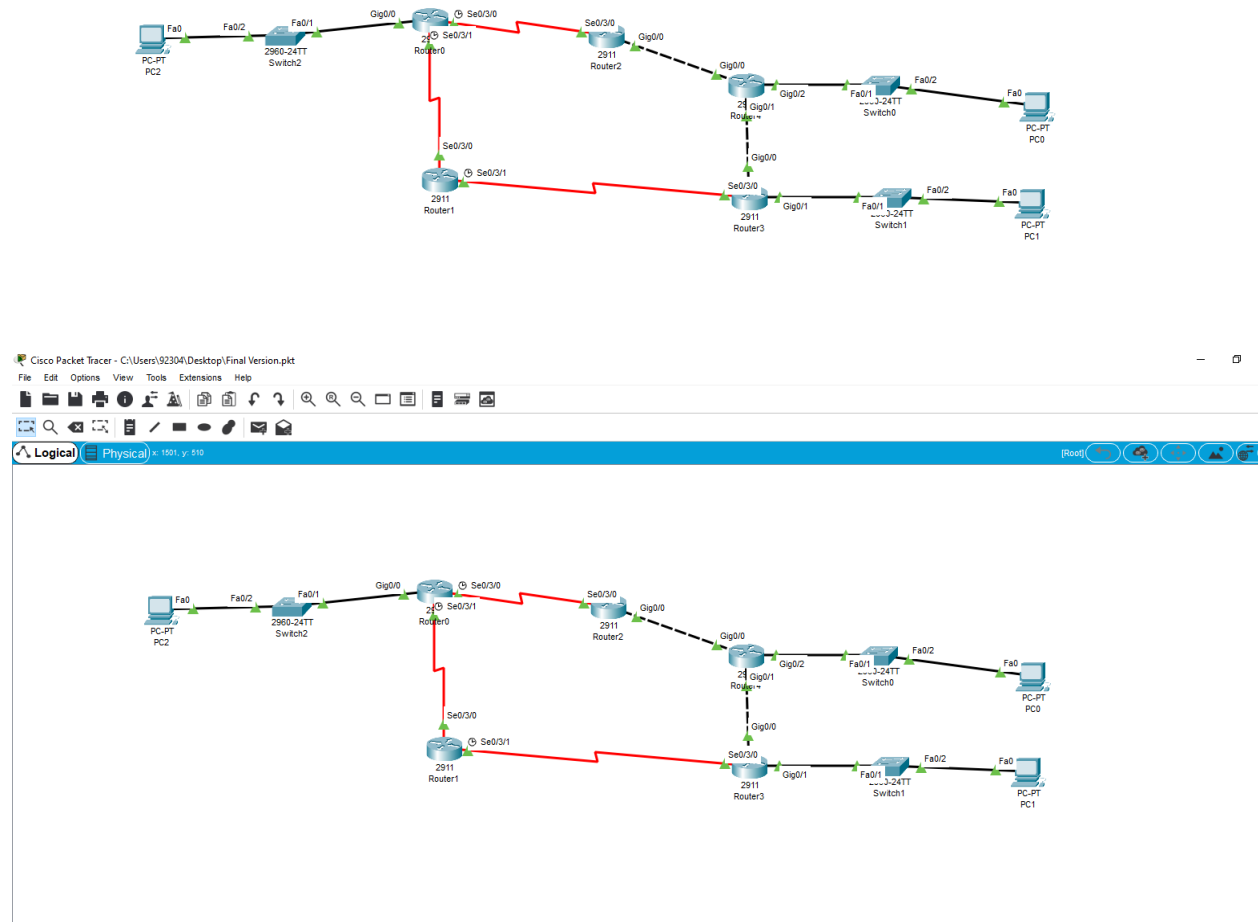
The IP address is composed of two fundamental components, namely the routing prefix and the subnet ID. These components serve to represent the global and local scopes of the IP address, respectively. The determination of the routing prefix, which refers to the division of bits, is frequently made by internet registries and internet service providers (ISPs). The initial 48 bits of an IPv6 address are commonly referred to as the site prefix and are positioned on the left-hand side of the address. The subsequent 16 bits represent the subnet identification. The topology of the site is determined by the subnet identifier. The interface ID, including the last 64 bits, can be configured using either automatic or manual means.

Types of IPv6 addresses

- It is worth noting that IPv6 does not have broadcast addresses, despite its support for a diverse range of address types and formats. The following are many prevalent formats for representing IPv6 addresses:
- The concept of global unicast. IP addresses that include the prefix "2001:" are capable of being routed. The public IP address serves as the IPv4 counterpart of global unicast addresses.
- The address in question is a unicast address. One method for determining the interface of a node.
- The term "Anycast identifier" refers to a unique identification used in the context of Anycast networking. In order to assign identifiers to a set of network interfaces that are distributed across many nodes.
- To clarify, a multicast IP address is being referred to. A unique identifier utilised for the purpose of multicasting. Multicasting enables the simultaneous transmission of a single packet to several recipients.
- Establishing connections between regional postal codes is a crucial task in the field of postal logistics. The device is not linked to the internet and possesses one of two distinct categories of internal addresses. The prefix group "fe80:" is utilised to identify link local addresses that are solely employed within a private network.
- Specifically, postal codes that are specific to a certain locality. This is an alternate variant of an internal address that is not routable on the internet.

IPV6 IMPLEMENTATION OF RIP IN PACKET TRACER

FINAL PACKET TRACER TOPOLOGY



CONFIGURATION STEPS

1. Configure IPv6 address in router1 interfaces
2. Configure IPv6 address in router2 interfaces
3. Configure IPv6 address in router3 interfaces
4. Configure IPv6 address in router4 interfaces
5. Configure IPv6 address in router5 interfaces
6. Configure IPv6 address in PCs
7. Check IP configuration in routers

8. Configure Rip IPV6 in all routers for routing
9. Verify configuration by and test network by ping result

IP ADDRESSING

From	To	IPv6 address / Network
Router 0	Router 1	2001:a::1/64
Router 0	Router 2	2001:b::1/64
Router 1	Router 3	2001:c::1/64
Router 2	Router 4	2001:d::1/64
Router 3	Router 4	2001:e::1/64
LAN 1	3001:a::1/64	
LAN 2	3001:b::1/64	
LAN 3	3001:c::1/64	

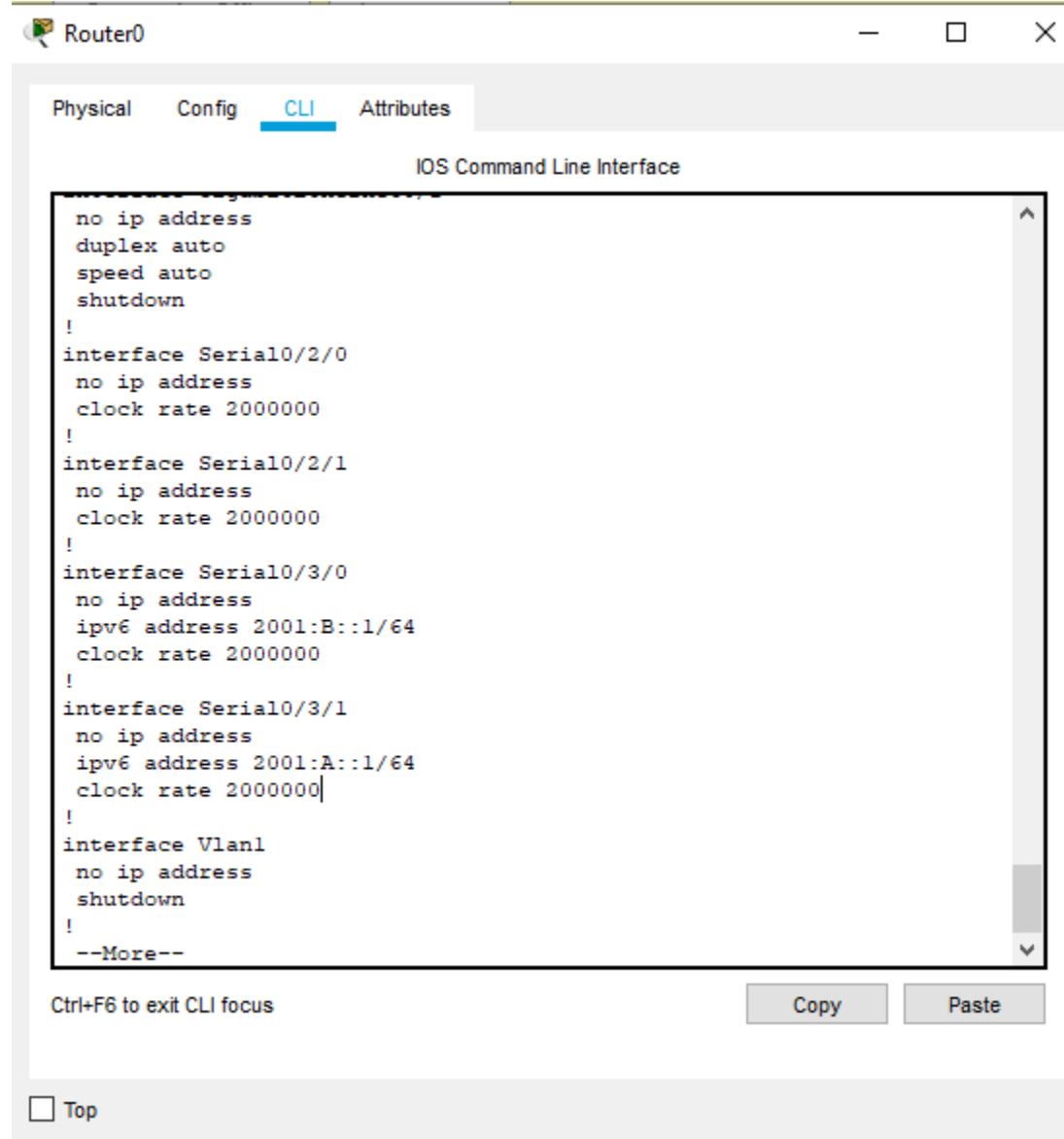
Command to add the Ipv6 address to interface is:-

Enabling IPv6 on the interfaces is recommended subsequent to its global activation. In order to activate IPv6 on a specific interface, the "ipv6 enable" command is utilised.

The EUI-64 format is employed in the generation of an IPv6 Global Unicast Address. The aforementioned format has been previously deliberated upon. The interface identifier of the entire IPv6 address is generated using the MAC address in this particular format. The newly generated interface identifier is subsequently appended to the network identifier.

The command "ip address ipv6-address/prefix-length eui-64" is utilised to configure an interface using the Extended Unique Identifier (EUI-64) format. The IPv6 address assigned to us is 2001:AAAA:BBBB:CCCC::/64, and the prefix length remains unchanged. Upon configuring IPv6, the aforementioned address and MAC address will be utilised in the derivation of the ultimate EUI-64 Global Unicast Address.

ROUTER 0:



The screenshot shows a window titled "Router0" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there are four tabs: "Physical", "Config", "CLI" (which is selected and highlighted in blue), and "Attributes". The "CLI" tab displays the "IOS Command Line Interface" with the following configuration text:

```
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
clock rate 2000000
!
interface Serial0/3/0
no ip address
ipv6 address 2001:B::1/64
clock rate 2000000
!
interface Serial0/3/1
no ip address
ipv6 address 2001:A::1/64
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
--More--
```

Below the CLI window, there is a status bar that says "Ctrl+F6 to exit CLI focus". To the right of this status bar are two buttons: "Copy" and "Paste". At the bottom left of the window, there is a checkbox labeled "Top" which is currently unchecked.

ROUTER 1

Physical Config CLI Attributes

IOS Command Line Interface

```
interface Serial0/2/0
  no ip address
  clock rate 2000000
!
interface Serial0/2/1
  no ip address
  clock rate 2000000
!
interface Serial0/3/0
  no ip address
  ipv6 address 2001:A::2/64
!
interface Serial0/3/1
  no ip address
  ipv6 address 2001:C::1/64
  clock rate 2000000
!
interface Vlan1
  no ip address
  shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
--More--
```

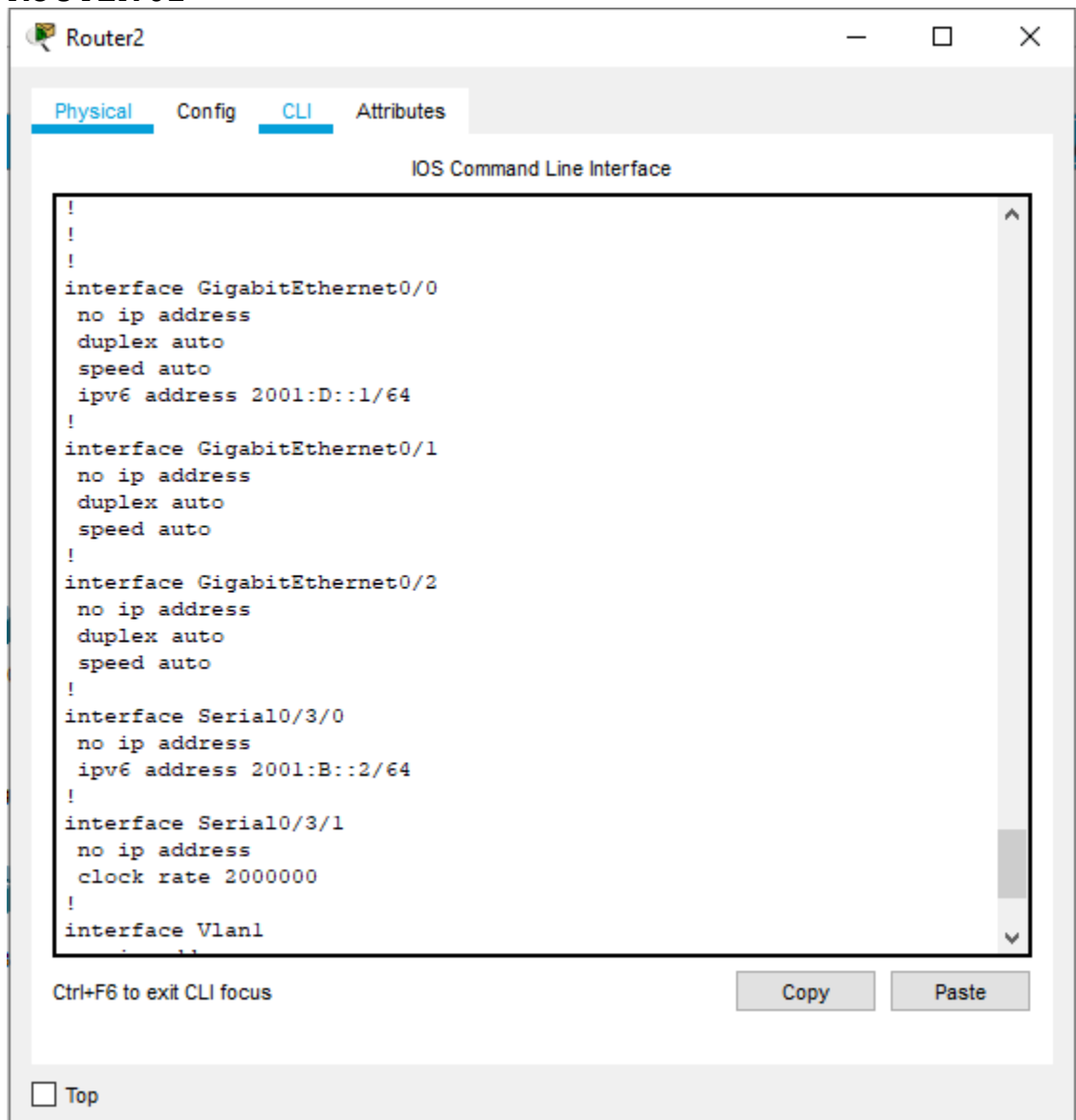
Ctrl+F6 to exit CLI focus

Copy

Paste

☐ Top

ROUTER 02



ROUTER 03

 Router3 — □ ×

Physical

Config

CLI

Attributes

IOS Command Line Interface

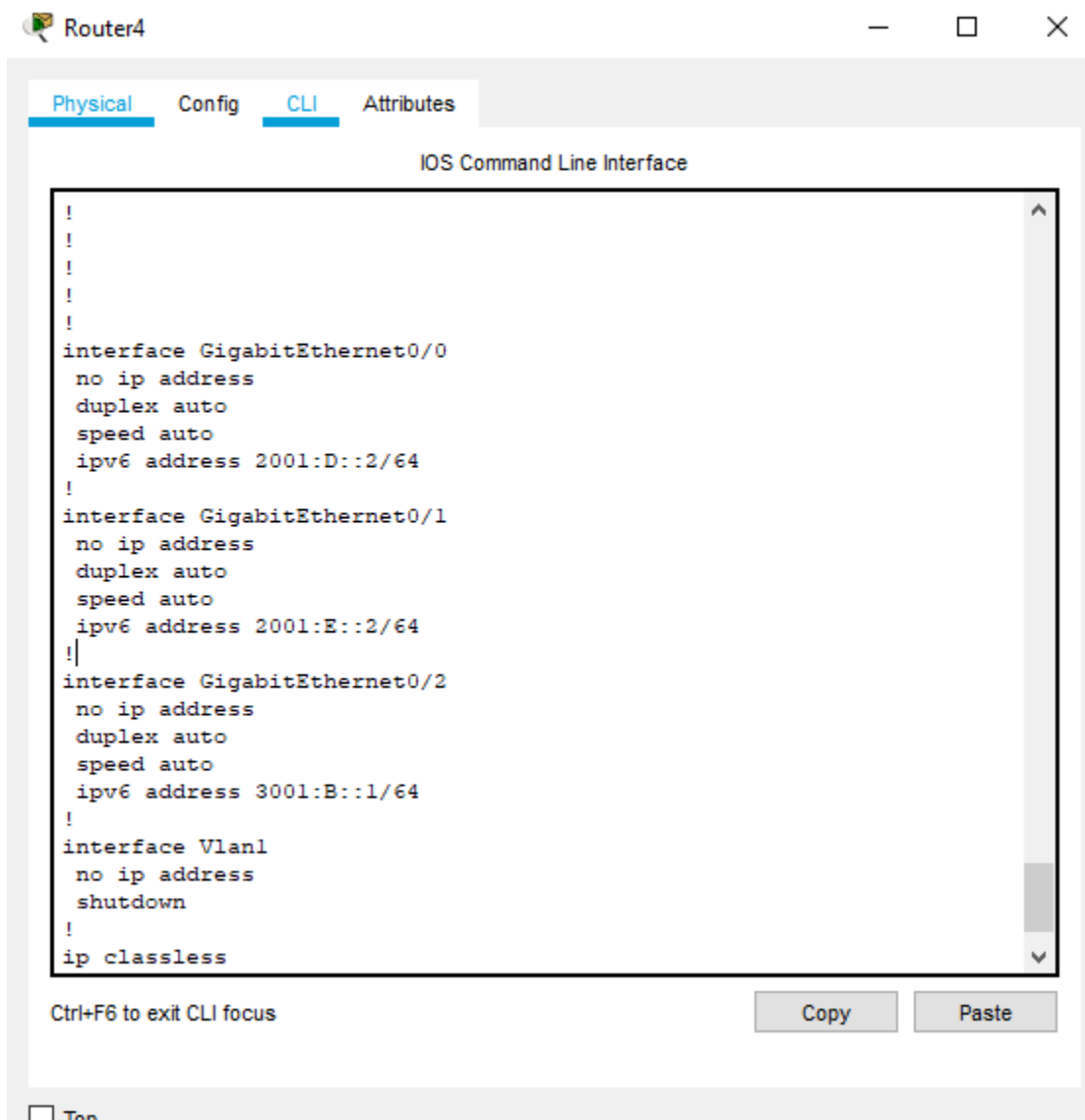
```
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
  ipv6 address 2001:E::1/64
!
interface GigabitEthernet0/1
  no ip address
  duplex auto
  speed auto
  ipv6 address 3001:C::1/64
!
interface GigabitEthernet0/2
  no ip address
  duplex auto
  speed auto
!
interface Serial0/3/0
  no ip address
  ipv6 address 2001:C::2/64
!
```

Ctrl+F6 to exit CLI focus

Copy

Paste

ROUTER 4



IPV6 CONFIGURATION IN LAN PCS

LAN – 01 PC 01

IPv6 Address: 3001:a::2 / 64

Gateway Address: 3001:a::1

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address 3001:A::2 / 64

Link Local Address FE80::20C:CFFF:FE0D:C60

IPv6 Gateway 3001:A::1

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

LAN – 02 PC 01

IPv6 Address: 3001:b::2 / 64

Gateway Address: 3001:a::1

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address 3001:B::2 / 64

Link Local Address FE80::260:3EFF:FE08:3462

IPv6 Gateway 3001:B::1

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

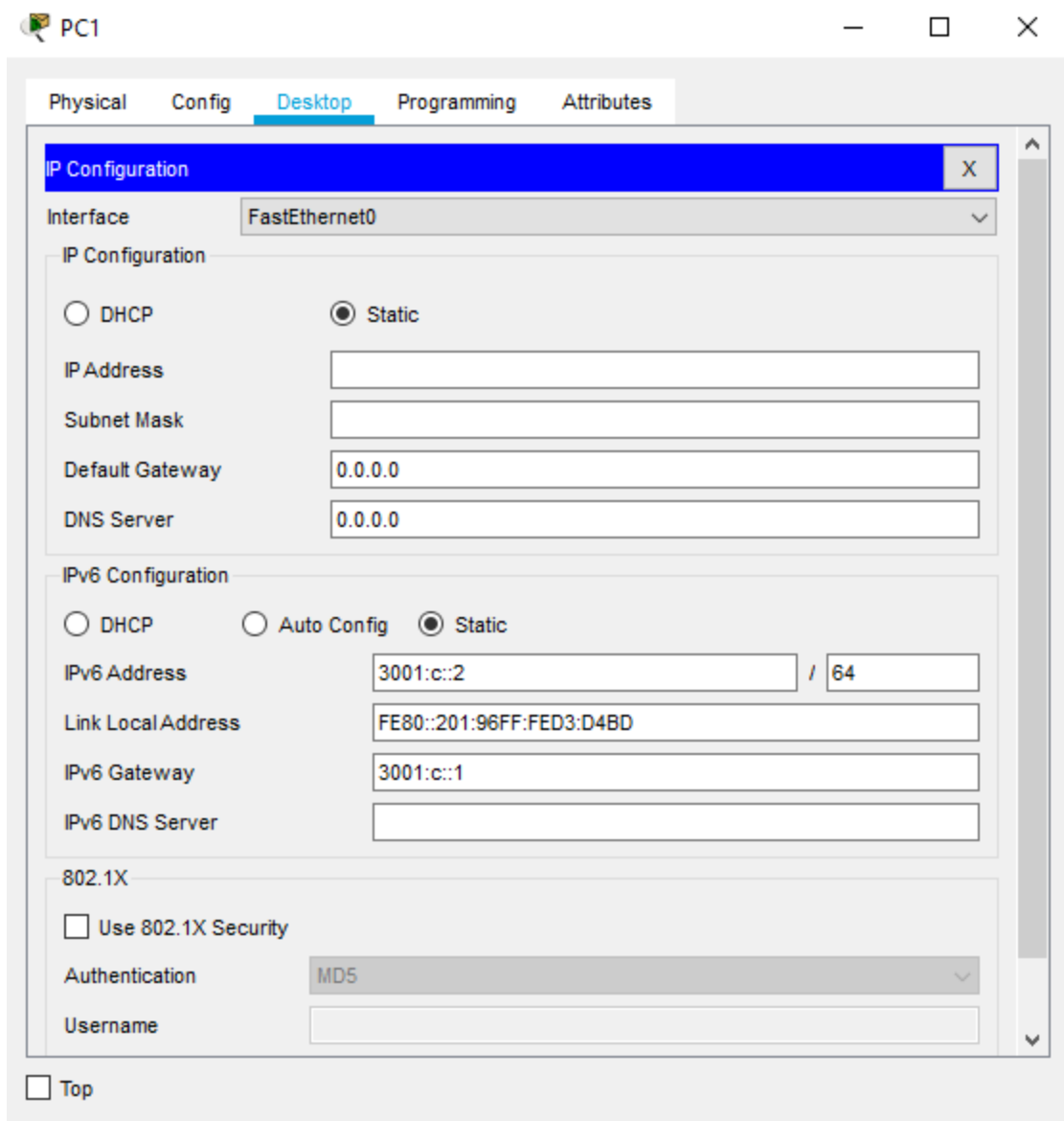
Username

☐ Top

LAN – 03 PC 01

IPv6 Address: 3001:c::2 / 64

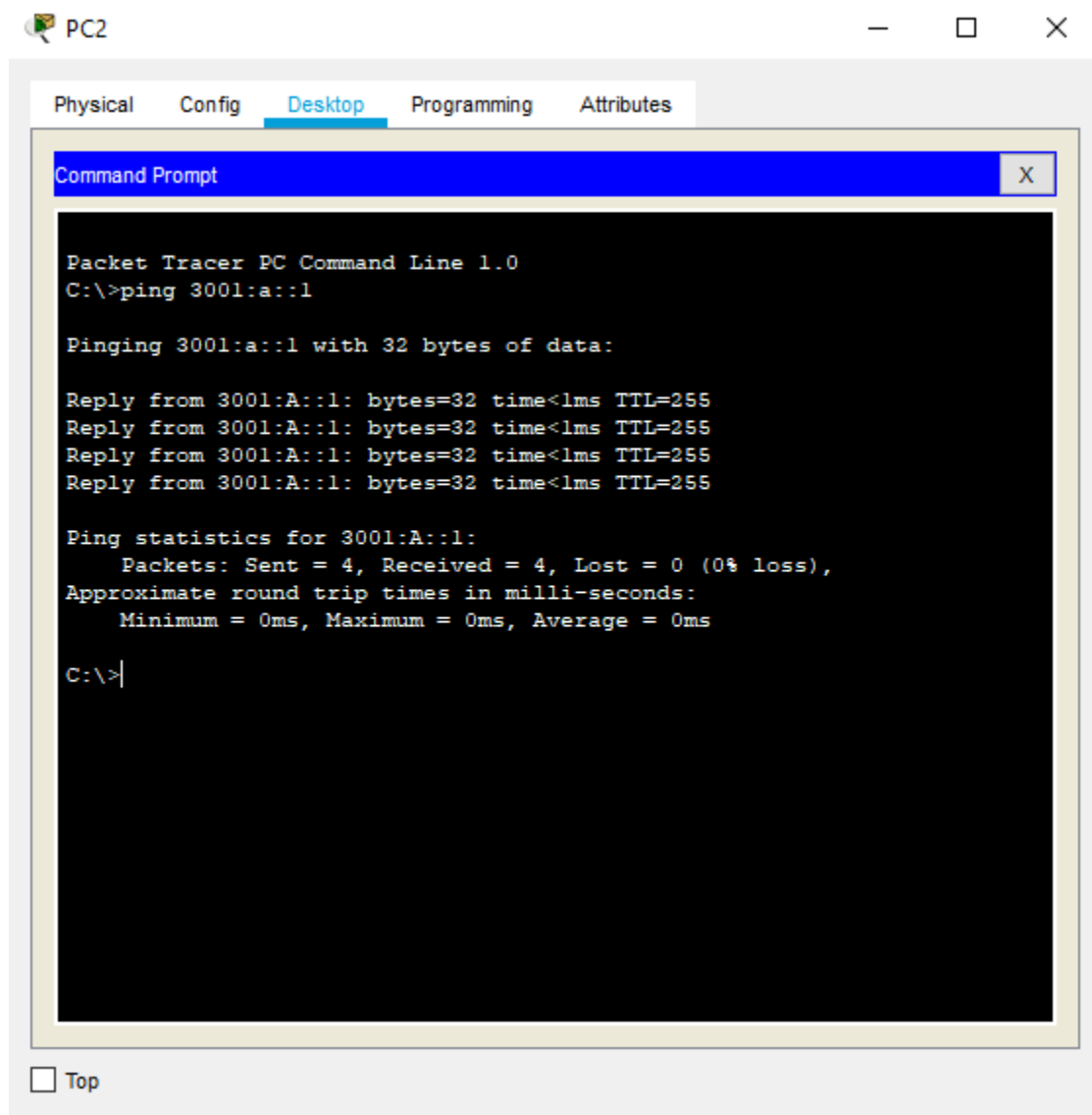
Gateway Address: 3001:a::1



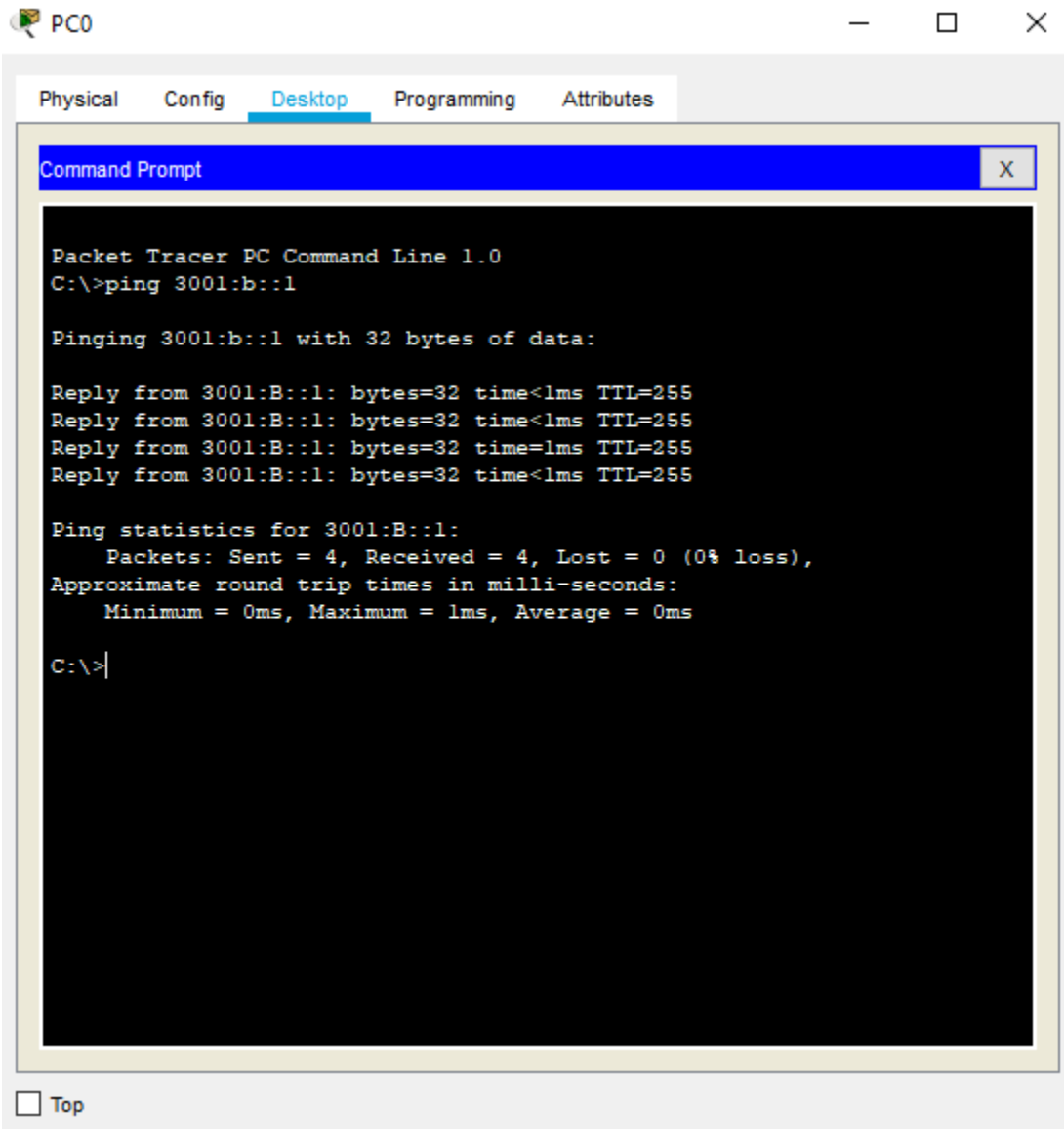
PING RESULTS FROM PCS TO LOCAL GATEWAY

The purpose of utilizing the ping command is to ascertain the presence of connectivity between two interconnected nodes. The notion of IPv6 Ping is analogous to that of IPv4 Ping, with the key distinction being the use of IPv6. The format of IPv6 Ping exhibits several variations when compared to IPv4 Ping. Two significant discrepancies may be observed between IP address and keyword forms. In order to utilize IPv6 Ping, it is necessary to input the command "ping ipv6" followed by the desired IPv6 address for communication.

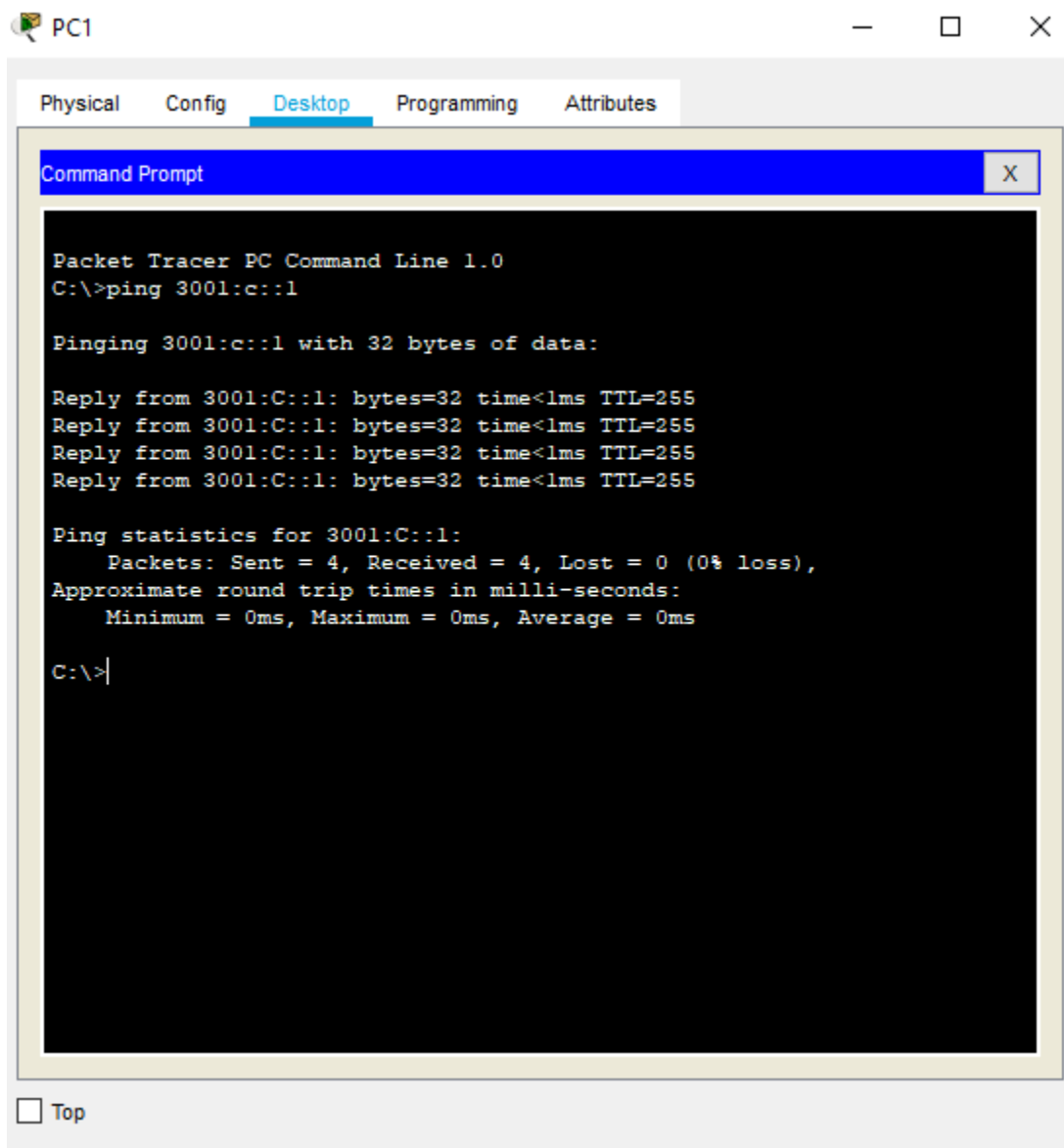
LAN - 01 PC



LAN -02 PC



LAN -03 PC



CONFIGURATION OF RIP IPV6 IN ROUTERS

In order to activate IPv6 on a Cisco router, the "ipv6 unicast-routing" command is employed subsequent to accessing configuration mode using the "configure terminal" command. This command, provided by Cisco, enables the activation of IPv6 on the whole router. This procedure can be implemented prior to configuring any IPv6 routing protocol or interface settings.

 Router0

Physical Config CLI Attributes

```
!  
!  
!  
no ip cef  
ipv6 unicast-routing  
!  
no ipv6 cef  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX1524RWD0-  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
  no ip address  
  duplex auto  
  speed auto  
  ipv6 address 3001:A::1/64  
  ipv6 rip Cisco enable  
!
```

IOS Command Line Interface

```
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
  ipv6 address 2001:E::1/64
  ipv6 rip Cisco enable
!
interface GigabitEthernet0/1
  no ip address
  duplex auto
  speed auto
  ipv6 address 3001:C::1/64
  ipv6 rip Cisco enable
!
interface GigabitEthernet0/2
  no ip address
  duplex auto
  speed auto
!
interface Serial0/3/0
  no ip address
  ipv6 address 2001:C::2/64
  ipv6 rip Cisco enable
!
interface Serial0/3/1
  no ip address
  clock rate 2000000
--More--
```

Ctrl+F6 to exit CLI focus

Copy

F



Physical Config CLI Attributes

IOS Command Line Interface

```

!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
no ip address
duplex auto
speed auto
ipv6 address 2001:D::2/64
ipv6 rip Cisco enable
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
ipv6 address 2001:E::2/64
ipv6 rip Cisco enable
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
ipv6 address 3001:B::1/64
ipv6 rip Cisco enable
!
interface Vlan1
no ip address
shutdown
!
ipv6 router rip Cisco
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
--More--

```

Ctrl+F6 to exit CLI focus

Copy

 Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
  no ip address  
  duplex auto  
  speed auto  
  ipv6 address 2001:D::1/64  
  ipv6 rip Cisco enable  
!  
interface GigabitEthernet0/1  
  no ip address  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/2  
  no ip address  
  duplex auto  
  speed auto  
!  
interface Serial0/3/0  
  no ip address  
  ipv6 address 2001:B::2/64  
  ipv6 rip Cisco enable  
!  
interface Serial0/3/1  
  no ip address  
  clock rate 2000000  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
ipv6 router rip Cisco  
!  
ip classless  
!  
ip flow-export version 9  
!  
!  
!  
!  
!  
--More--
```

Ctrl+F6 to exit CLI focus

Copy

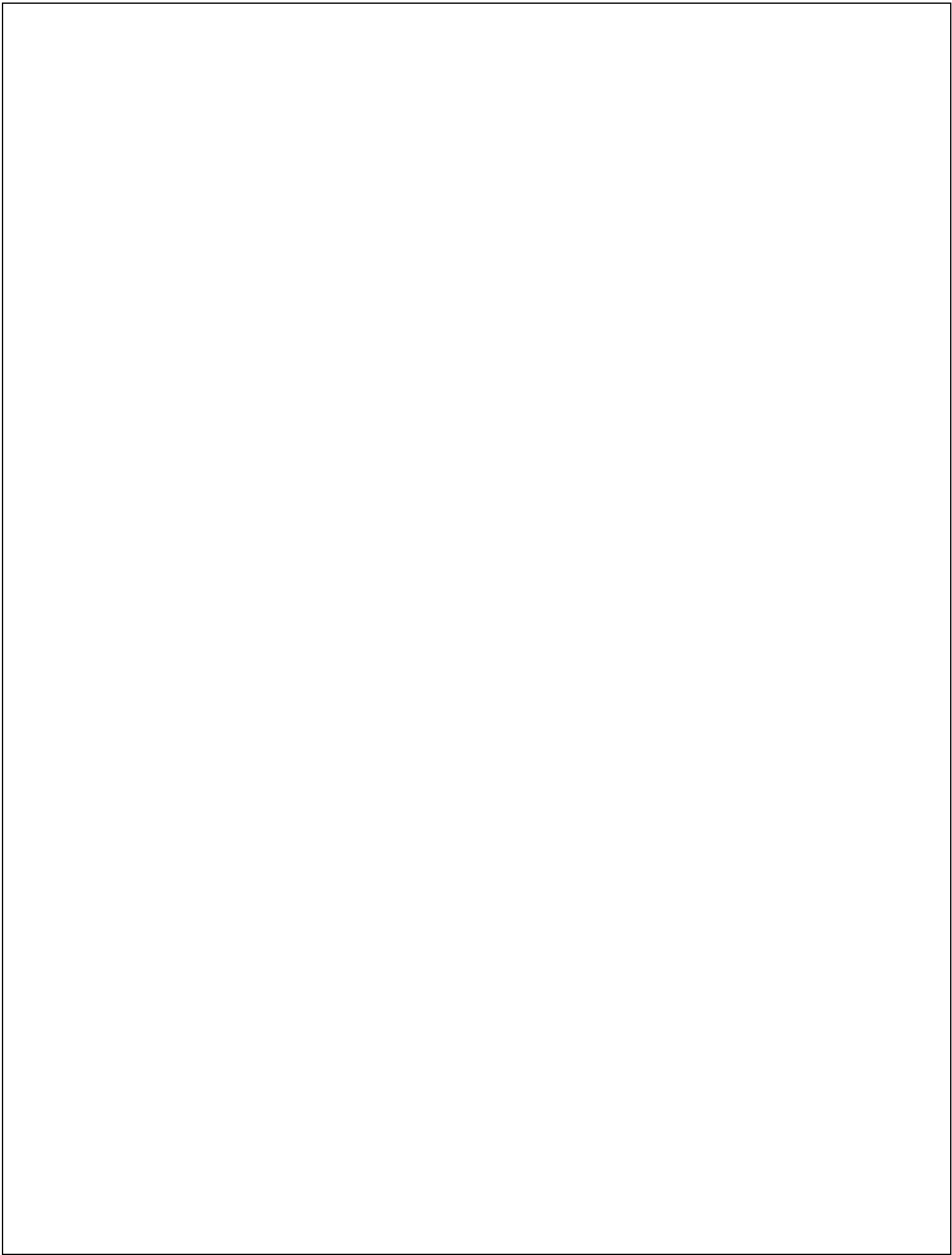
☐ Top

IOS Command Line Interface

```
speed auto
shutdown
!
interface GigabitEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
clock rate 2000000
!
interface Serial0/3/0
no ip address
ipv6 address 2001:A::2/64
ipv6 rip Cisco enable
!
interface Serial0/3/1
no ip address
ipv6 address 2001:C::1/64
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ipv6 router rip Cisco
!
ip classless
!
ip flow-export version 9
!
!
!
--More--
```

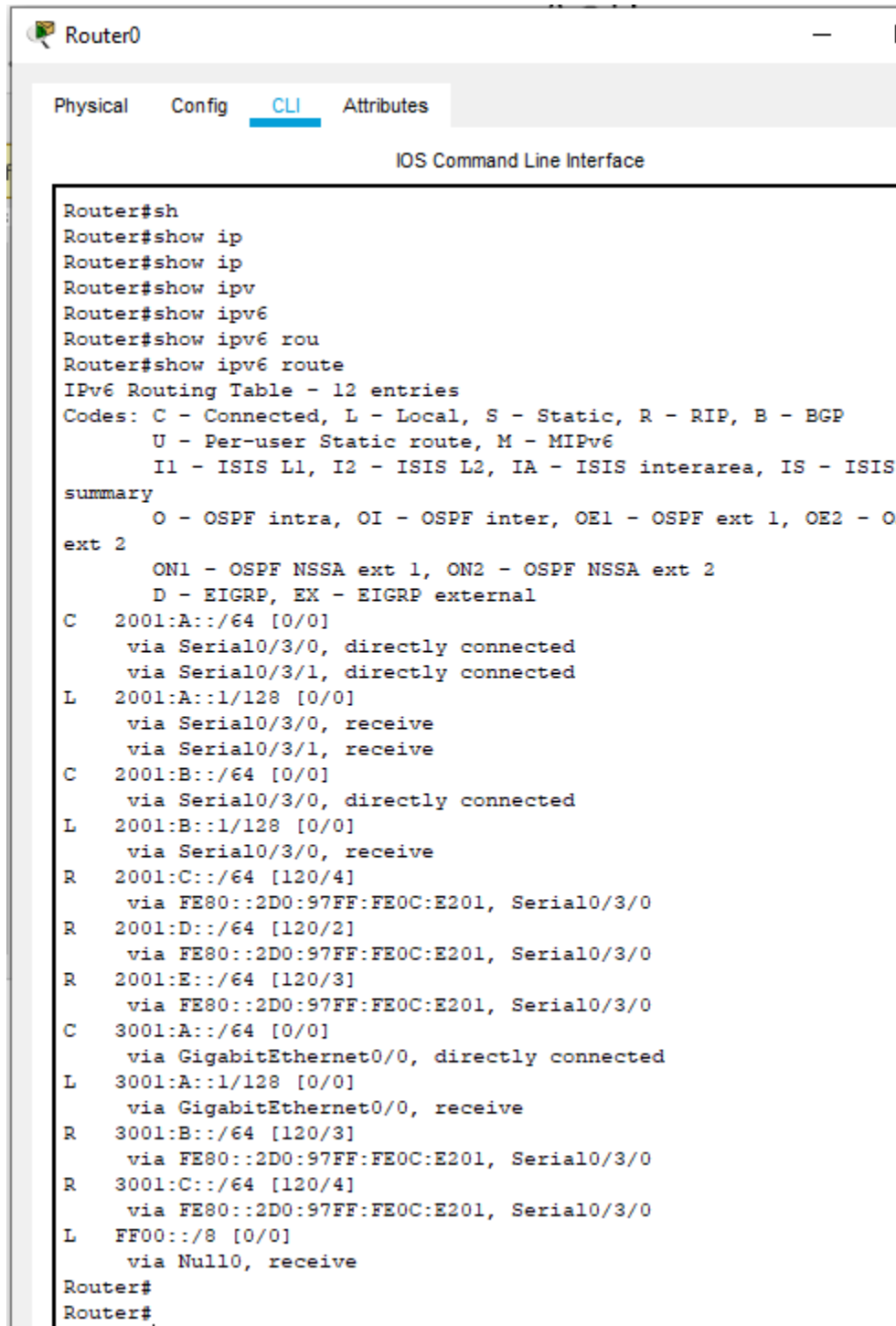
Ctrl+F6 to exit CLI focus

Copy



IPV6 ROUTES VERIFICATION IN ROUTERS

ROUTER 0



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#sh
Router#show ip
Router#show ip
Router#show ipv
Router#show ipv6
Router#show ipv6 rou
Router#show ipv6 route
IPv6 Routing Table - 12 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - O
ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
C   2001:A::/64 [0/0]
    via Serial0/3/0, directly connected
    via Serial0/3/1, directly connected
L   2001:A::1/128 [0/0]
    via Serial0/3/0, receive
    via Serial0/3/1, receive
C   2001:B::/64 [0/0]
    via Serial0/3/0, directly connected
L   2001:B::1/128 [0/0]
    via Serial0/3/0, receive
R   2001:C::/64 [120/4]
    via FE80::2D0:97FF:FE0C:E201, Serial0/3/0
R   2001:D::/64 [120/2]
    via FE80::2D0:97FF:FE0C:E201, Serial0/3/0
R   2001:E::/64 [120/3]
    via FE80::2D0:97FF:FE0C:E201, Serial0/3/0
C   3001:A::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L   3001:A::1/128 [0/0]
    via GigabitEthernet0/0, receive
R   3001:B::/64 [120/3]
    via FE80::2D0:97FF:FE0C:E201, Serial0/3/0
R   3001:C::/64 [120/4]
    via FE80::2D0:97FF:FE0C:E201, Serial0/3/0
L   FF00::/8 [0/0]
    via Null0, receive
Router#
Router#
```

Router 1

Router1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
ip flow-export version 9
!
!
!
!
!

Router#
Router#
Router#
Router#sh
Router#show ipv
Router#show ipv6 ro
Router#show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF
ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:A::/64 [0/0]
    via Serial0/3/0, directly connected
L   2001:A::2/128 [0/0]
    via Serial0/3/0, receive
R   2001:B::/64 [120/2]
    via FE80::210:11FF:FEC0:E101, Serial0/3/0
C   2001:C::/64 [0/0]
    via Serial0/3/1, directly connected
L   2001:C::1/128 [0/0]
    via Serial0/3/1, receive
R   2001:D::/64 [120/3]
    via FE80::210:11FF:FEC0:E101, Serial0/3/0
R   2001:E::/64 [120/4]
    via FE80::210:11FF:FEC0:E101, Serial0/3/0
R   3001:A::/64 [120/2]
    via FE80::210:11FF:FEC0:E101, Serial0/3/0
R   3001:B::/64 [120/4]
    via FE80::210:11FF:FEC0:E101, Serial0/3/0
R   3001:C::/64 [120/5]
    via FE80::210:11FF:FEC0:E101, Serial0/3/0
L   FF00::/8 [0/0]
    via Null0, receive
Router#
```

Ctrl+F6 to exit CLI focus

CopyPaste

ROUTER 4

 Router4

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
!
!
!
!
!

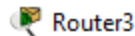
Router#
Router#
Router#
Router#sh
Router#show ipv
Router#show ipv6 rou
Router#show ipv6 route
IPv6 Routing Table - 12 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS
summary
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF
ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
R   2001:A::/64 [120/3]
    via FE80::2D0:97FF:FE0C:E201, GigabitEthernet0/0
R   2001:B::/64 [120/2]
    via FE80::2D0:97FF:FE0C:E201, GigabitEthernet0/0
R   2001:C::/64 [120/2]
    via FE80::202:4AFF:FE39:4901, GigabitEthernet0/1
C   2001:D::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L   2001:D::2/128 [0/0]
    via GigabitEthernet0/0, receive
C   2001:E::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L   2001:E::2/128 [0/0]
    via GigabitEthernet0/1, receive
R   3001:A::/64 [120/3]
    via FE80::2D0:97FF:FE0C:E201, GigabitEthernet0/0
C   3001:B::/64 [0/0]
    via GigabitEthernet0/2, directly connected
L   3001:B::1/128 [0/0]
    via GigabitEthernet0/2, receive
R   3001:C::/64 [120/2]
    via FE80::202:4AFF:FE39:4901, GigabitEthernet0/1
L   FF00::/8 [0/0]
    via Null0, receive
Router#
Router#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

ROUTER 3

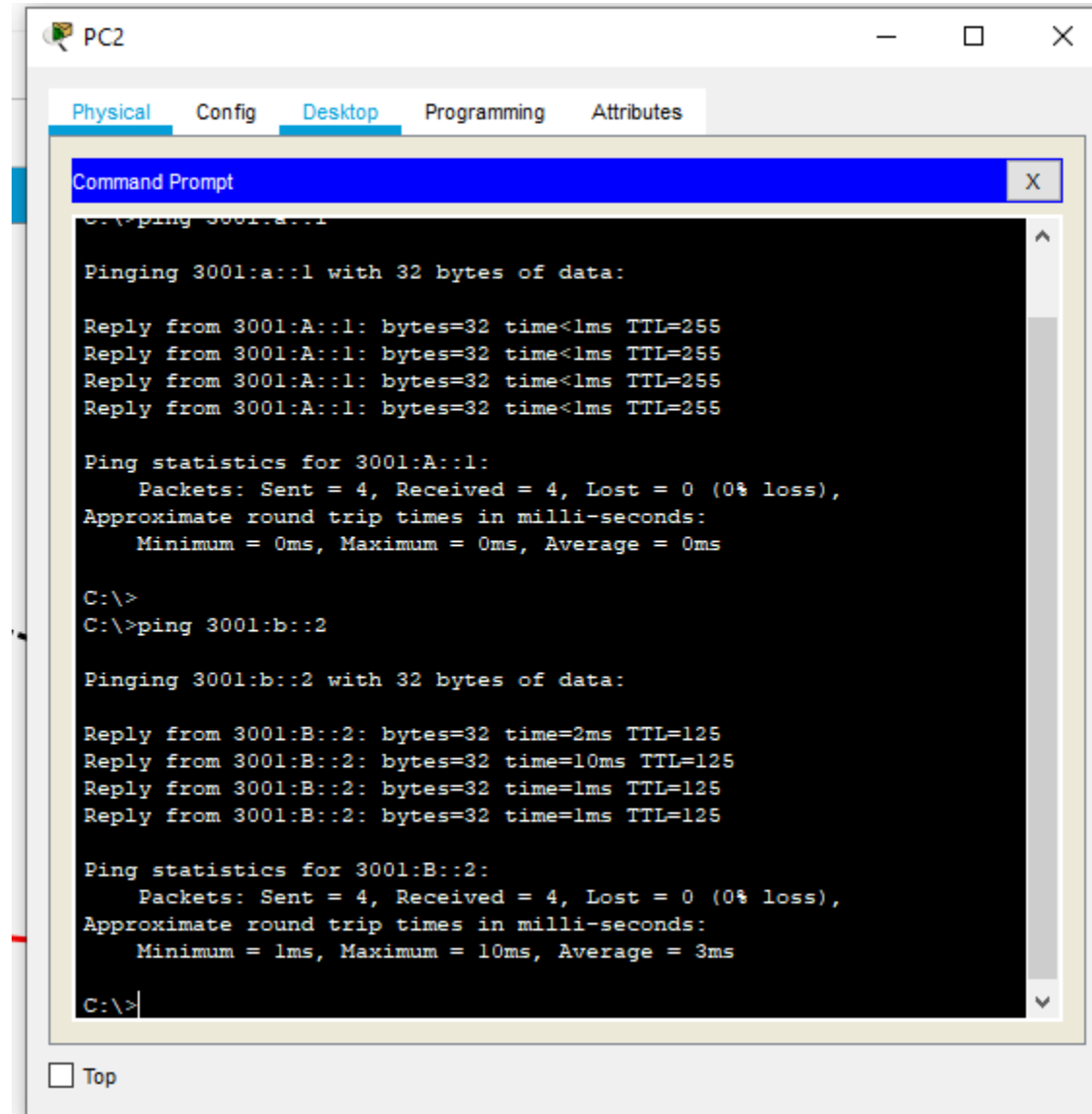


Physical Config CLI Attributes

```
Router#
Router#
Router#sh
Router#show ipv
Router#show ipv6 ro
Router#show ipv6 route
IPv6 Routing Table - 12 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
R   2001:A::/64 [120/4]
    via FE80::20A:41FF:FE80:5E02, GigabitEthernet0/0
R   2001:B::/64 [120/3]
    via FE80::20A:41FF:FE80:5E02, GigabitEthernet0/0
C   2001:C::/64 [0/0]
    via Serial0/3/0, directly connected
L   2001:C::2/128 [0/0]
    via Serial0/3/0, receive
R   2001:D::/64 [120/2]
    via FE80::20A:41FF:FE80:5E02, GigabitEthernet0/0
C   2001:E::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L   2001:E::1/128 [0/0]
    via GigabitEthernet0/0, receive
R   3001:A::/64 [120/4]
    via FE80::20A:41FF:FE80:5E02, GigabitEthernet0/0
R   3001:B::/64 [120/2]
    via FE80::20A:41FF:FE80:5E02, GigabitEthernet0/0
C   3001:C::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L   3001:C::1/128 [0/0]
    via GigabitEthernet0/1, receive
L   FF00::/8 [0/0]
    via Null0, receive
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
```

PING FINAL RESULT FROM LANS

LAN -1 TO LAN 2 PING



The screenshot shows a window titled "PC2" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the execution of two ping commands. The first command is "C:\>ping 3001:a::1", which results in four successful replies from 3001:A::1 with 32 bytes of data, each taking less than 1ms and having a TTL of 255. The statistics for this ping show 4 packets sent and received, 0% loss, and 0ms round trip times. The second command is "C:\>ping 3001:b::2", which results in four successful replies from 3001:B::2 with 32 bytes of data. The round trip times are 2ms, 10ms, 1ms, and 1ms, with a TTL of 125. The statistics for this ping show 4 packets sent and received, 0% loss, and an average round trip time of 3ms. The Command Prompt window has a "Top" button at the bottom left.

```
C:\>ping 3001:a::1

Pinging 3001:a::1 with 32 bytes of data:

Reply from 3001:A::1: bytes=32 time<1ms TTL=255
Reply from 3001:A::1: bytes=32 time<1ms TTL=255
Reply from 3001:A::1: bytes=32 time<1ms TTL=255
Reply from 3001:A::1: bytes=32 time<1ms TTL=255

Ping statistics for 3001:A::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
C:\>ping 3001:b::2

Pinging 3001:b::2 with 32 bytes of data:

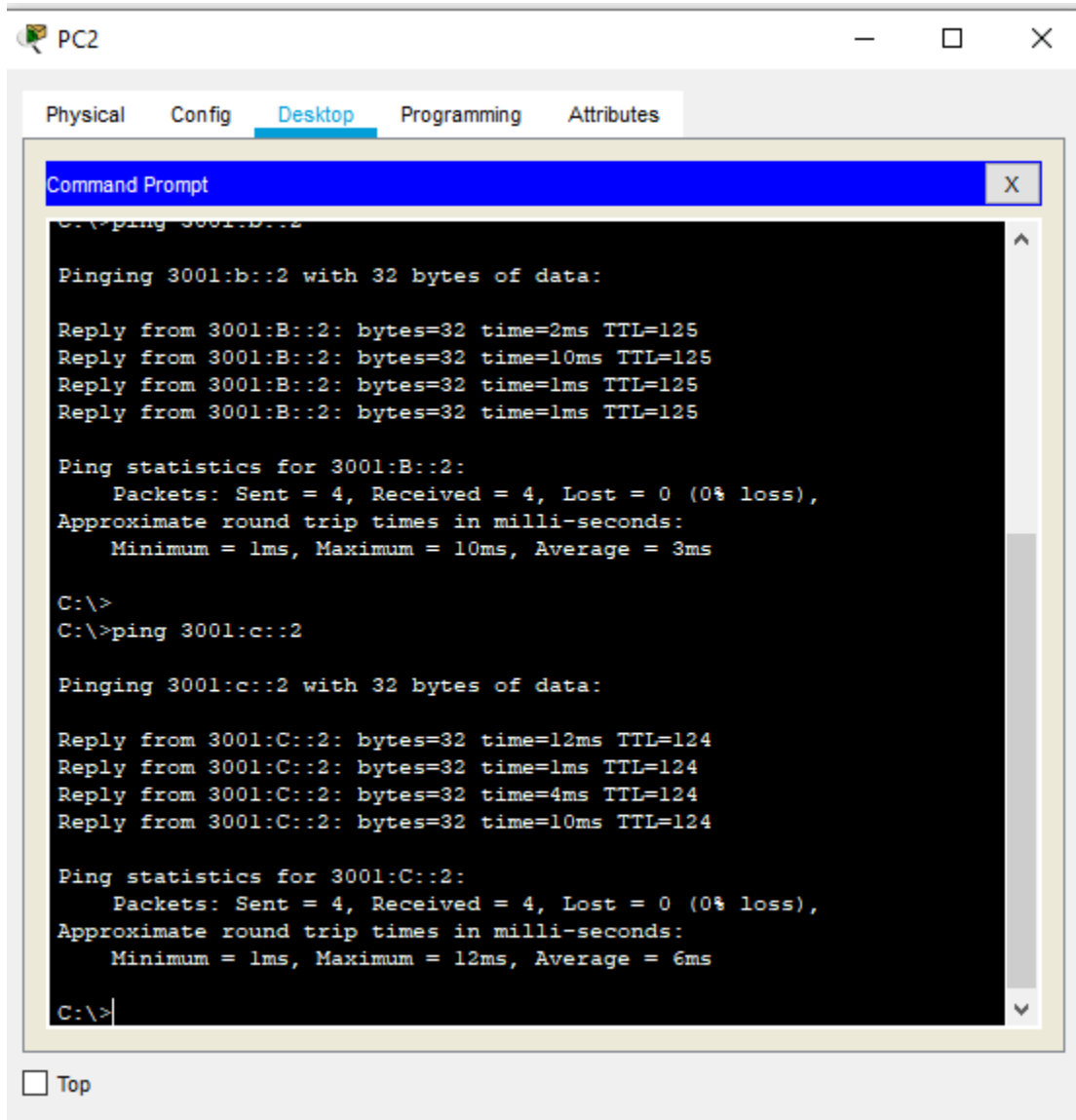
Reply from 3001:B::2: bytes=32 time=2ms TTL=125
Reply from 3001:B::2: bytes=32 time=10ms TTL=125
Reply from 3001:B::2: bytes=32 time=1ms TTL=125
Reply from 3001:B::2: bytes=32 time=1ms TTL=125

Ping statistics for 3001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 3ms

C:\>
```

☐ Top

LAN -1 TO LAN 3



The screenshot shows a window titled "PC2" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the execution of two ping commands. The first command is "C:\>ping 3001:B::2", which results in four successful replies from 3001:B::2 with varying times (2ms, 10ms, 1ms, 1ms) and a TTL of 125. The statistics show 4 packets sent, 4 received, 0 lost, and an average round trip time of 3ms. The second command is "C:\>ping 3001:C::2", which results in four successful replies from 3001:C::2 with varying times (12ms, 1ms, 4ms, 10ms) and a TTL of 124. The statistics show 4 packets sent, 4 received, 0 lost, and an average round trip time of 6ms. The Command Prompt window has a blue title bar and a scroll bar on the right. Below the Command Prompt window, there is a checkbox labeled "Top".

```
C:\>ping 3001:B::2

Pinging 3001:b::2 with 32 bytes of data:

Reply from 3001:B::2: bytes=32 time=2ms TTL=125
Reply from 3001:B::2: bytes=32 time=10ms TTL=125
Reply from 3001:B::2: bytes=32 time=1ms TTL=125
Reply from 3001:B::2: bytes=32 time=1ms TTL=125

Ping statistics for 3001:B::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 3ms

C:\>
C:\>ping 3001:c::2

Pinging 3001:c::2 with 32 bytes of data:

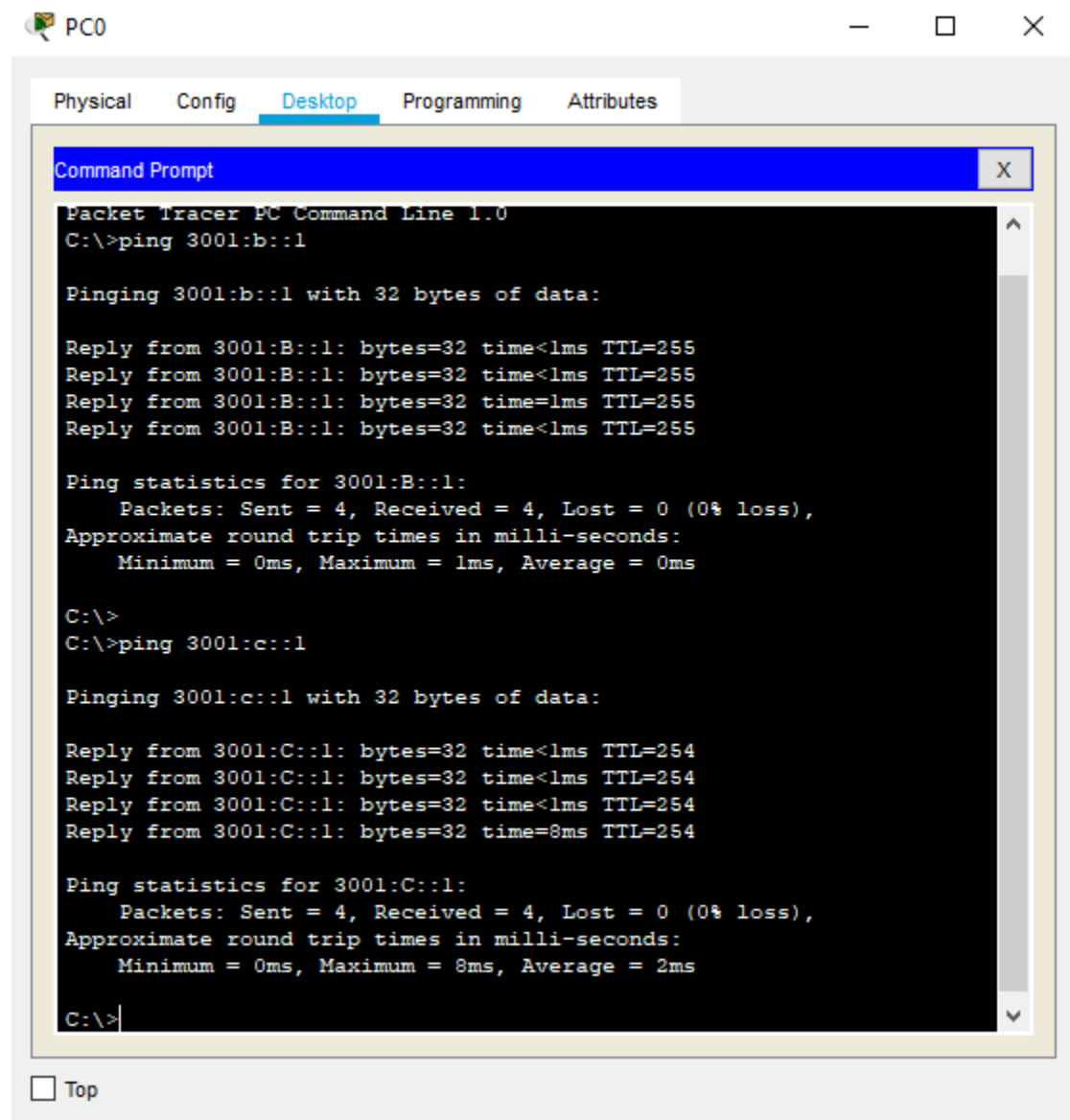
Reply from 3001:C::2: bytes=32 time=12ms TTL=124
Reply from 3001:C::2: bytes=32 time=1ms TTL=124
Reply from 3001:C::2: bytes=32 time=4ms TTL=124
Reply from 3001:C::2: bytes=32 time=10ms TTL=124

Ping statistics for 3001:C::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 6ms

C:\>
```

☐ Top

LAN -2 TO LAN -3



OSPF IMPLEMENTATION WITH IPV6

The Open Shortest Path First (OSPF) link-state routing protocol employs an independent shortest path first algorithm to ascertain the most efficient route between the source and destination routers. The Open Shortest Path First (OSPF) protocol was developed by the Internet Engineering Task Force (IETF) with the objective of enhancing packet transmission within a substantial autonomous system or routing domain. The protocol in question operates on port 89 and utilizes the AD value 110. It is classified as a network layer protocol. The multicast address 224.0.0.5 is utilized for standard Open Shortest Path First (OSPF) communication, while the multicast address 224.0.0.6 is employed for updates related to the Designated Router (DR) and Backup Designated Router (BDR).

The IPv6 addressing semantics were eliminated from both the OSPF packets and the primary LSA types in order to achieve protocol independence inside the network. The following shifts have significant importance:

The only authority to own an individual's physical address is vested solely in an LSA. The inclusion of party addresses is omitted from the "hello" packet. The interface on the provided connection is assigned a distinct identification by the router from which it originates. This identifier is then saved in a designated field known as Interface ID.

The inclusion of network addresses in the router-LSA or network-LSA types is no longer observed. In contrast to an IP address and network mask, an LSA address is expressed in the form of a prefix followed by a length.

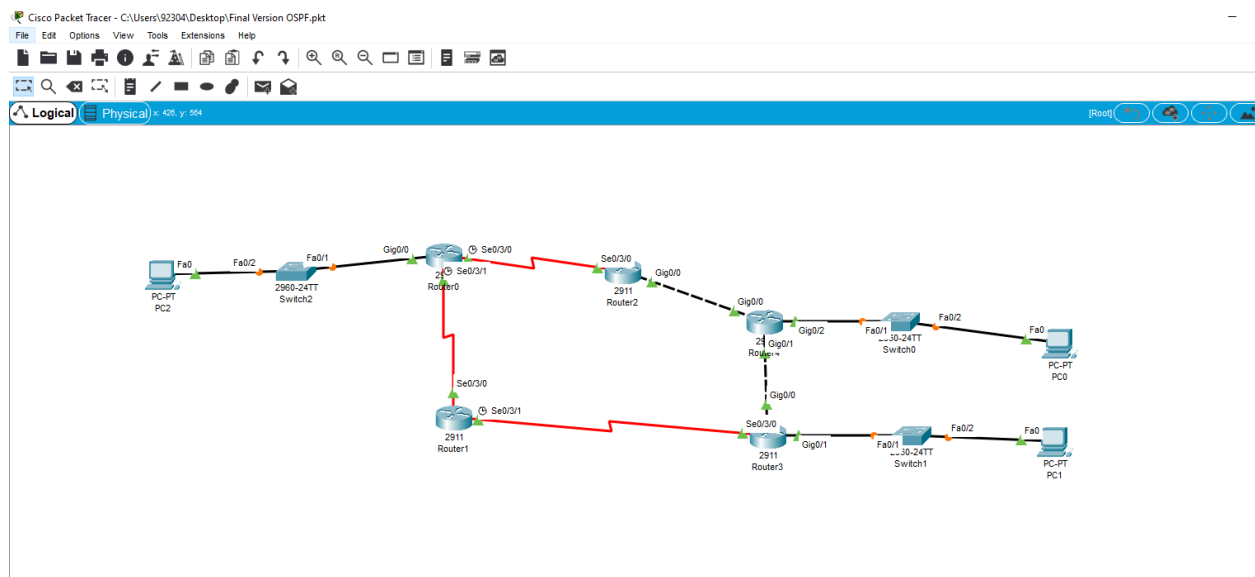
In the context of OSPF for IPv6, it is necessary for the router ID to serve as a unique identifier for each adjacent router. The individual is assigned a numerical IP address under the IPv4 protocol. IPv6 does not support peer-to-peer (P2P) connections.

IP ADDRESSING TABLE

From	To	IPv6 address / Network
Router 0	Router 1	2001:a::1/64
Router 0	Router 2	2001:b::1/64

Router 1	Router 3	2001:c::1/64
Router 2	Router 4	2001:d::1/64
Router 3	Router 4	2001:e::1/64
LAN 1	3001:a::1/64	
LAN 2	3001:b::1/64	
LAN 3	3001:c::1/64	

NETWORK TOPOLOGY



IMPLEMENTATION OF OSPF IPV6

Enable IPv6 unicast routing on both routers in global configuration mode :

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#
Router(config)#do sh ru
Building configuration...

Current configuration : 1120 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
```

 Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
```

 Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524505Q-
!
!
!
```

 Router4

Physical Config CLI Attributes

IOS Command Line Interface

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
no ip cef
ipv6 unicast-routing
!
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524EQ10-
!
!
!
```

IP ADDRESS CONFIGURATION IN INTERFACES

ROUTER 0:

 Router0 — □ ×

Physical Config **CLI** Attributes

IOS Command Line Interface

```
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/2/0
no ip address
clock rate 2000000
!
interface Serial0/2/1
no ip address
clock rate 2000000
!
interface Serial0/3/0
no ip address
ipv6 address 2001:B::1/64
clock rate 2000000
!
interface Serial0/3/1
no ip address
ipv6 address 2001:A::1/64
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
--More--
```

Ctrl+F6 to exit CLI focus Copy Paste

☐ Top

ROUTER 1

 Router1

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
interface Serial10/2/0
  no ip address
  clock rate 2000000
!
interface Serial10/2/1
  no ip address
  clock rate 2000000
!
interface Serial10/3/0
  no ip address
  ipv6 address 2001:A::2/64
!
interface Serial10/3/1
  no ip address
  ipv6 address 2001:C::1/64
  clock rate 2000000
!
interface Vlan1
  no ip address
  shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
--More--
```

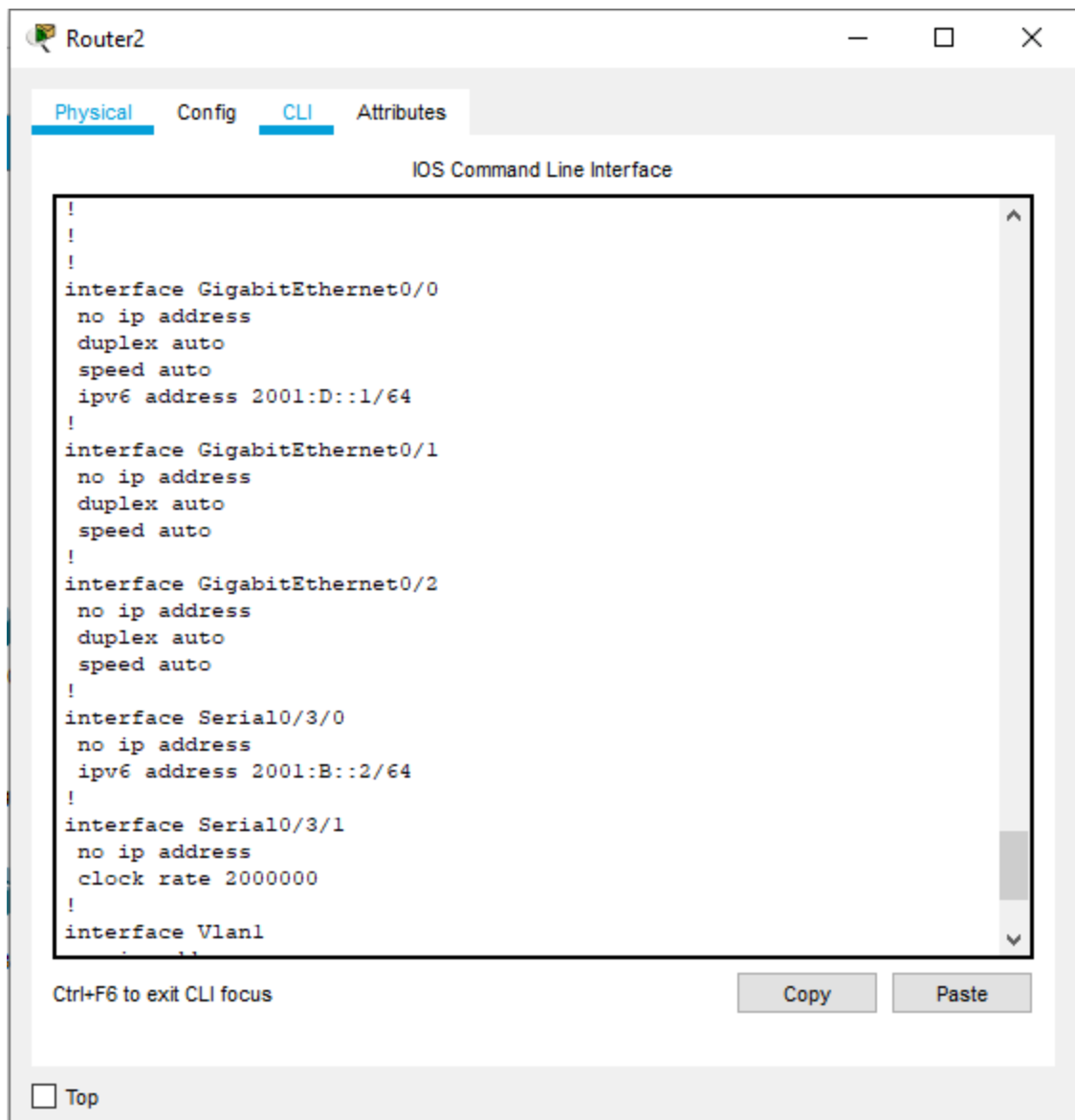
Ctrl+F6 to exit CLI focus

Copy

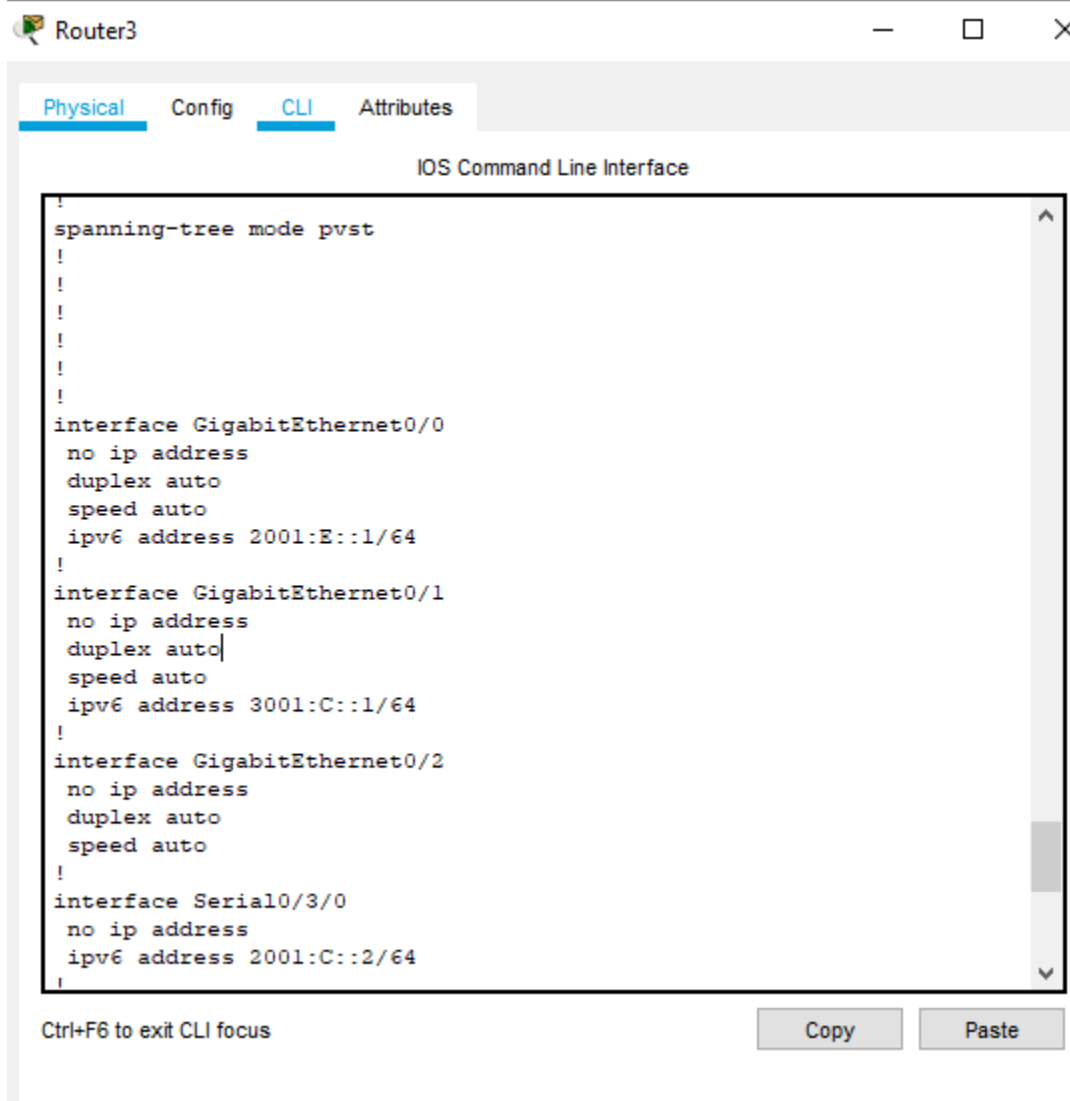
Paste

☐ Top

ROUTER 02



ROUTER 03

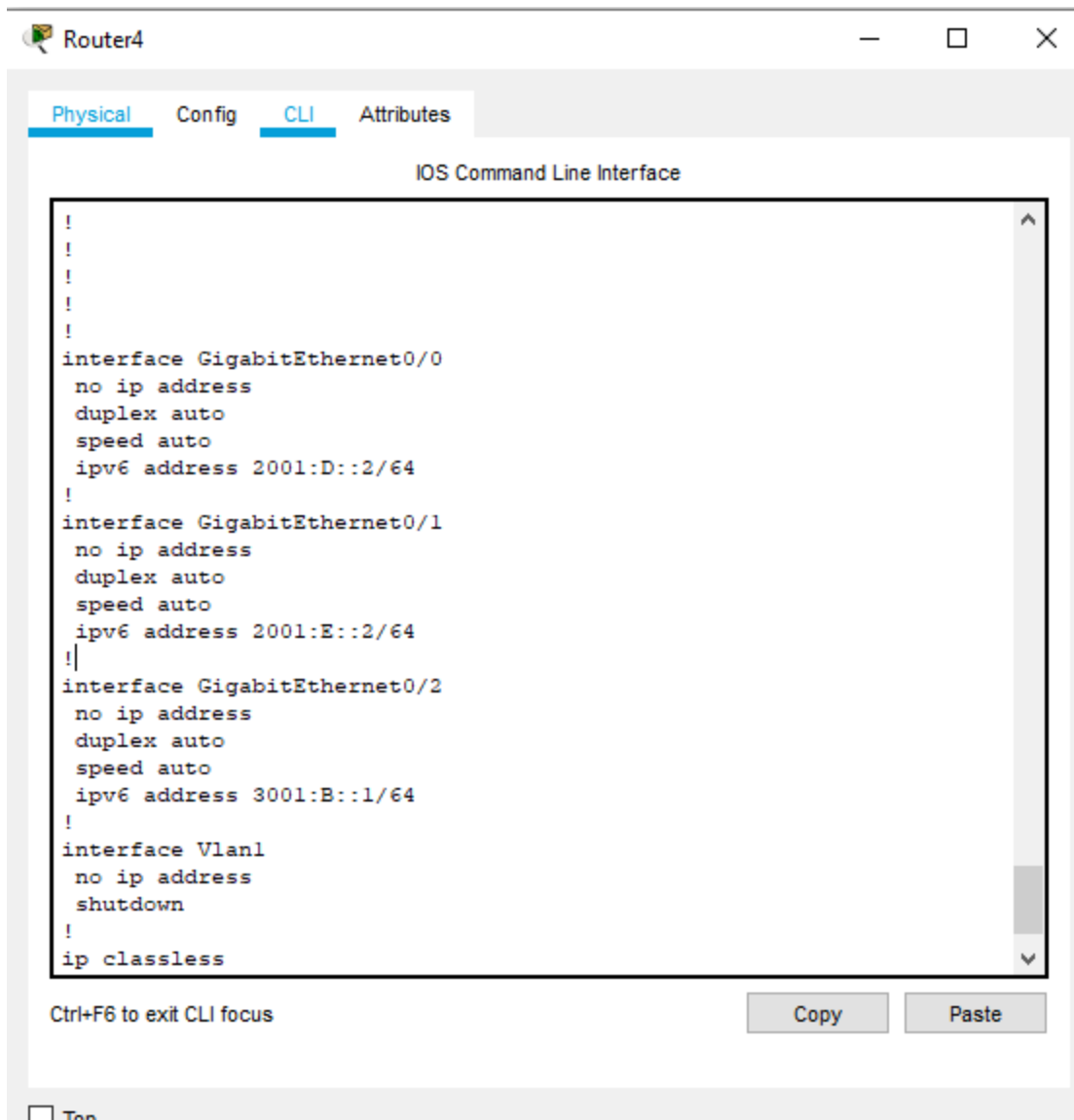


The screenshot shows a window titled "Router3" with a standard Windows-style title bar (minimize, maximize, close buttons). Below the title bar is a tabbed interface with four tabs: "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is currently selected and highlighted in blue. The main content area of the "CLI" tab is titled "IOS Command Line Interface" and contains a text editor with the following configuration commands:

```
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
  no ip address
  duplex auto
  speed auto
  ipv6 address 2001:E::1/64
!
interface GigabitEthernet0/1
  no ip address
  duplex auto
  speed auto
  ipv6 address 3001:C::1/64
!
interface GigabitEthernet0/2
  no ip address
  duplex auto
  speed auto
!
interface Serial0/3/0
  no ip address
  ipv6 address 2001:C::2/64
!
```

At the bottom of the window, there is a status bar with the text "Ctrl+F6 to exit CLI focus" on the left and two buttons, "Copy" and "Paste", on the right.

ROUTER 4



IPV6 CONFIGURATION IN LAN PCS

LAN – 01 PC 01

IPv6 Address: 3001:a::2 / 64

Gateway Address: 3001:a::1

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address 3001:A::2 / 64

Link Local Address FE80::20C:CFFF:FE0D:C60

IPv6 Gateway 3001:A::1

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

LAN - 02 PC 01

IPv6 Address: 3001:b::2 / 64

Gateway Address: 3001:a::1

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address:

Subnet Mask:

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address: 3001:B::2 / 64

Link Local Address: FE80::260:3EFF:FE08:3462

IPv6 Gateway: 3001:B::1

IPv6 DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

☐ Top

LAN – 03 PC 01

IPv6 Address: 3001:c::2 / 64

Gateway Address: 3001:a::1

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IP Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address 3001:c::2 / 64

Link Local Address FE80::201:96FF:FED3:D4BD

IPv6 Gateway 3001:c::1

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

☐ Top

OSPF COMMANDS

 Router0

Physical Config CLI Attributes

```
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
no ip address  
duplex auto  
speed auto  
ipv6 address 3001:A::1/64  
ipv6 ospf 10 area 0  
!  
interface GigabitEthernet0/1  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Serial0/2/0  
no ip address  
clock rate 2000000  
!  
interface Serial0/2/1  
no ip address  
clock rate 2000000  
!  
interface Serial0/3/0  
no ip address  
ipv6 address 2001:B::1/64  
ipv6 ospf 10 area 0  
clock rate 2000000  
!  
interface Serial0/3/1  
no ip address  
ipv6 address 2001:A::1/64  
ipv6 ospf 10 area 0  
clock rate 2000000  
!
```

Ctrl+F6 to exit CLI focus

Physical Config CLI Attributes

```

speed auto
shutdown
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Serial10/2/0
no ip address
clock rate 2000000
!
interface Serial10/2/1
no ip address
clock rate 2000000
!
interface Serial10/3/0
no ip address
ipv6 address 2001:A::2/64
ipv6 ospf 10 area 0
!
interface Serial10/3/1
no ip address
ipv6 address 2001:C::1/64
ipv6 ospf 10 area 0
clock rate 2000000
!
interface Vlan1
no ip address
shutdown
!
ipv6 router ospf 10
router-id 2.2.2.2
log-adjacency-changes
!
ipv6 router rip Cisco
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
--More--

```


Router2

Physical Config CLI Attributes

```
!  
!  
!  
!  
interface GigabitEthernet0/0  
  no ip address  
  duplex auto  
  speed auto  
  ipv6 address 2001:D::1/64  
  ipv6 ospf 10 area 0  
!  
interface GigabitEthernet0/1  
  no ip address  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/2  
  no ip address  
  duplex auto  
  speed auto  
!  
interface Serial0/3/0  
  no ip address  
  ipv6 address 2001:B::2/64  
  ipv6 ospf 10 area 0  
!  
interface Serial0/3/1  
  no ip address  
  clock rate 2000000  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
ipv6 router ospf 10  
  router-id 3.3.3.3  
  log-adjacency-changes  
!  
ipv6 router rip Cisco  
!  
ip classless  
!  
ip flow-export version 9  
!  
!  
!  
!  
--More--
```

Ctrl+F6 to exit CLI focus

OSPF Peer Verification

 Router0

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
inspect          CBAC (Context Based Access Control) information
interface        IPv6 interface status and configuration
nat              IPv6 NAT-PT information
neighbors        Show IPv6 neighbor cache entries
ospf             OSPF information
protocols        IPv6 Routing Protocols
rip              RIP routing protocol status
route            Show IPv6 route table entries
Router#show ipv6 ops
Router#show ipv6 osp
Router#show ipv6 ospf ?
<1-65535>        Process ID number
border-routers   Border and Boundary Router Information
database         Database summary
interface        Interface information
neighbor         Neighbor list
virtual-links    Virtual link information
<cr>
Router#show ipv6 ospf nei
Router#show ipv6 ospf neighbor

Neighbor ID      Pri   State           Dead Time   Interface ID
Interface
3.3.3.3          0    FULL/ -         00:00:31   4
Serial0/3/0
2.2.2.2          0    FULL/ -         00:00:33   6
Serial0/3/1
Router#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

☐ Top



Router1



Physical

Config

CLI

Attributes

IOS Command Line Interface

```
Router(config)#  
Router(config)#  
Router(config)#  
Router(config)#  
Router(config)#exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#  
Router#  
Router#  
Router#  
Router#sh  
Router#show os  
Router#show osp  
Router#show ip  
Router#show ipv  
Router#show ipv6 osp  
Router#show ipv6 ospf nei  
Router#show ipv6 ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Interface ID
Interface				
1.1.1.1	0	FULL/ -	00:00:33	7
Serial0/3/0				
5.5.5.5	0	FULL/ -	00:00:30	4
Serial0/3/1				

```
Router#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

☐ Top

IOS Command Line Interface

GigabitEthernet0/1 from LOADING to FULL, Loading Done

```
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
```

%SYS-5-CONFIG_I: Configured from console by console

```
Router#
Router#shj
Router#sh
Router#show ipv
Router#show ipv6 os
Router#show ipv6 ospf nei
Router#show ipv6 ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Interface ID
Interface				
5.5.5.5	1	FULL/BDR	00:00:38	1
GigabitEthernet0/1				
3.3.3.3	1	FULL/BDR	00:00:37	1
GigabitEthernet0/0				

```
Router#
```

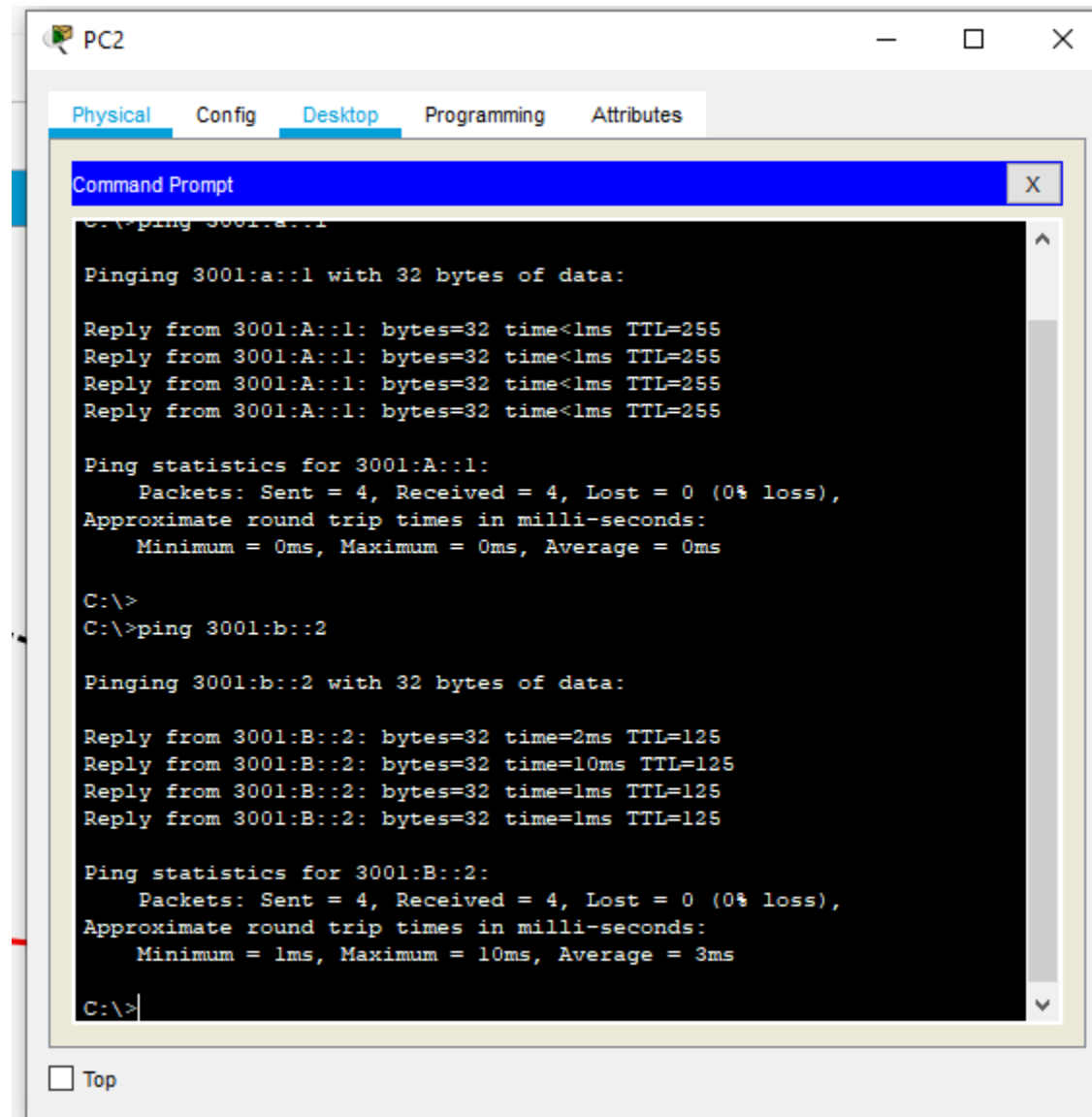
Ctrl+F6 to exit CLI focus

Copy

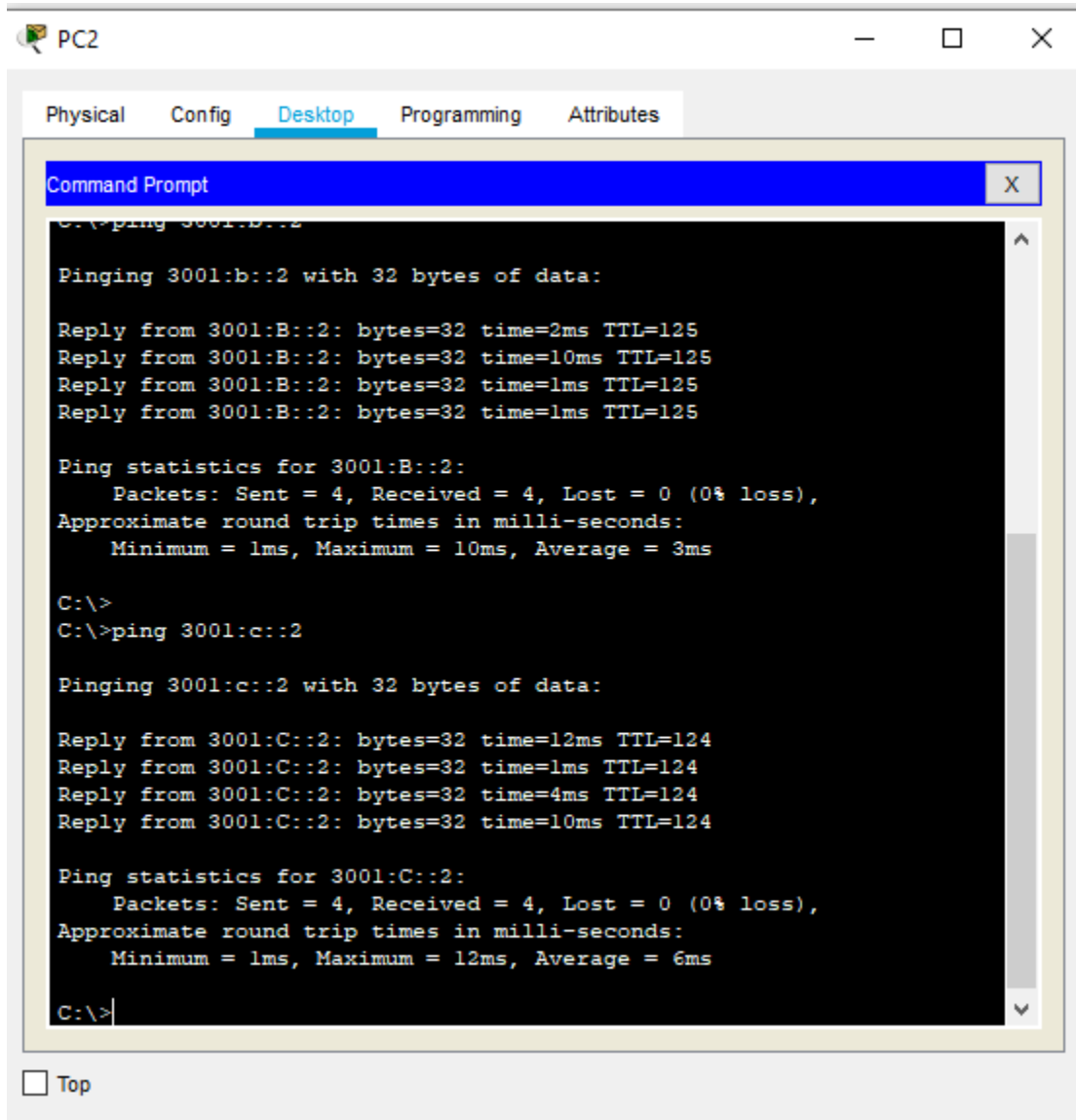
☐ Top

PING RESULT

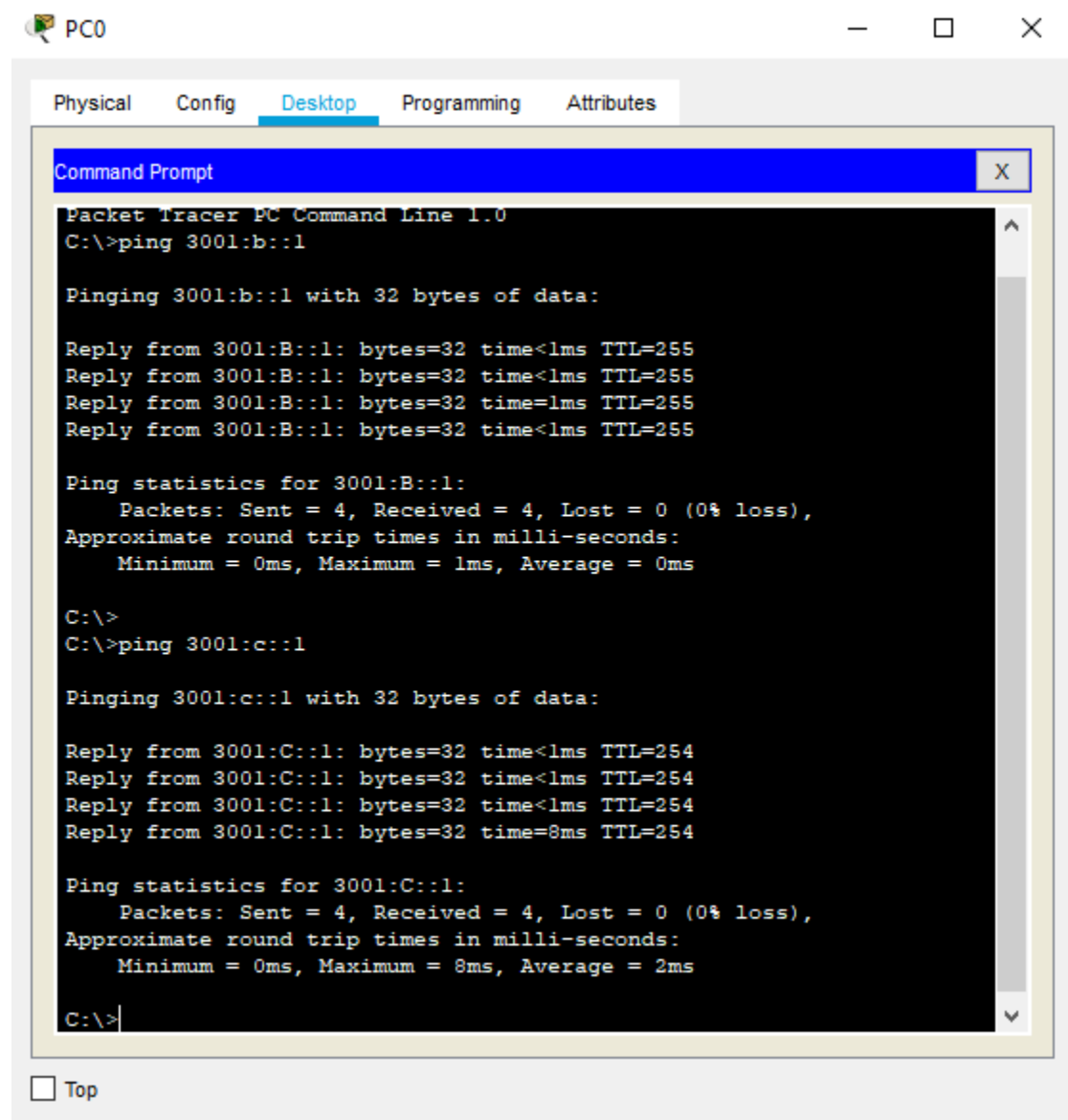
LAN -1 TO LAN 2 PING



LAN -1 TO LAN 3



LAN -2 TO LAN -3



```
Packet Tracer PC Command Line 1.0
C:\>ping 3001:b::1

Pinging 3001:b::1 with 32 bytes of data:

Reply from 3001:B::1: bytes=32 time<1ms TTL=255
Reply from 3001:B::1: bytes=32 time<1ms TTL=255
Reply from 3001:B::1: bytes=32 time=1ms TTL=255
Reply from 3001:B::1: bytes=32 time<1ms TTL=255

Ping statistics for 3001:B::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
C:\>ping 3001:c::1

Pinging 3001:c::1 with 32 bytes of data:

Reply from 3001:C::1: bytes=32 time<1ms TTL=254
Reply from 3001:C::1: bytes=32 time<1ms TTL=254
Reply from 3001:C::1: bytes=32 time<1ms TTL=254
Reply from 3001:C::1: bytes=32 time=8ms TTL=254

Ping statistics for 3001:C::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>
```

CONCLUSION

The acronym "RIPng" is an abbreviation for the Next Generation Routing Information Protocol. In order to mitigate any interference with the preexisting RIPv1 and RIPv2 protocols, the current protocol opts to employ port 521 as its designated transport protocol, UDP, as opposed to the conventional port 520. The need for an IPv6 iteration of the Routing Information Protocol (RIP) prompted the creation of RIPng, which exhibits compatibility with a diverse array of CISCO

routers as well as those manufactured by other manufacturers. The system is built upon the RIPv2 protocol, which is the second version of the Routing Information Protocol designed for Internet Protocol version 4. The final destination for multicast RIPv2 broadcasts is FF02::9, which corresponds to an IPv6 address. The functioning of RIPv2 has similarities to that of RIPv2 in several aspects.

The Open Shortest Path First (OSPF) link-state routing protocol employs an internal shortest path first algorithm to ascertain the most efficient route between the source and destination routers. The Open Shortest Path First (OSPF) protocol was developed by the Internet Engineering Task Force (IETF) with the aim of enabling efficient packet transmission inside a sizable autonomous system or routing domain. The protocol in question operates on port 89 and utilises the AD value of 110. It is classified as a network layer protocol. The multicast address 224.0.0.5 is utilised for standard Open Shortest Path First (OSPF) communication, while the multicast address 224.0.0.6 is employed for updates related to the Designated Router (DR) and Backup Designated Router (BDR).

OSPFv3 is the most recent version of the original Open Shortest Path First (OSPF) protocol. The system provides support for both the IPv4 and IPv6 address families. The protocol number assigned to the system is 89. Enables many occurrences inside a single connection. The creation of new Link State Advertisement (LSA) kinds is required in order to facilitate the installation of IPv6 prefixes.

A novel link state type field has been included to facilitate the identification of the scope of Link State Advertisement (LSA) flooding.

Managing Unidentified Categories in Latent Semantic Analysis

There are certain limitations associated with OSPFv3 that should be considered.

Lack of compatibility with legacy systems. OSPFv3 exhibits incompatibility with preceding iterations of the protocol.

Authentication without neighbouring entities is achieved by the utilisation of IPsec header extensions.

The presence of a concealed router identification is absent. In order to dynamically determine the Router-ID, OSPFv3 utilises the same mechanism as OSPFv2. However, it is necessary for human

intervention to configure the RID. If there are no IPv4 interfaces available, the Router-ID is configured as 0.0.0.0, hence preventing the establishment of an adjacency in OSPF.

Non-Broadcast Multiple Access (NBMA) systems: The discovery of neighbours is not inherently facilitated by Non-Broadcast Multiple Access (NBMA) connections.